

```
face_distances = face_recognition.face_distance(known_face_encodings, face_encoding)
best_match_index = np.argmax(face_distances)
if matches[best_match_index]:
    name = known_face_names[best_match_index]
    print((sensor.get_ambient()+3))
    print((sensor.get_object_1()+3))
    temp=(str ((sensor.get_object_1()+3))+"°C")
    bodytemp=(sensor.get_object_1() +3)
    if previous != name:
        previous=name
        print(name)
        #espeak.synth("Hey I Can Identify your face")
        #espeak.synth("Your name is")
        #espeak.synth(name)
        espeak.set_voice("En")
        #espeak.set_voice("whisper")
        espeak.synth("Hey hello ")
        espeak.synth(name)
        espeak.synth("your body temperature is ")
        espeak.synth(temp)
```

Fig. 4: Code for checking the temperature

```
#espeak.synth(name)
espeak.set_voice("En")
#espeak.set_voice("whisper")
espeak.synth("Hey hello ")
espeak.synth(name)
espeak.synth("your body temperature is ")
espeak.synth(temp)

if bodytemp < 37:
    espeak.synth("you can enter")
    imagestring=str(datetime.datetime.now().strftime("%I:%M:%S%p on %B %d,%Y"))

    text_file=open("entry.txt","a+")
    n= text_file.write(name)
    n= text_file.write(temp)
    n= text_file.write(datetime.datetime.now().strftime("%I:%M:%S%p on %B %d,%Y"))
    n= text_file.write("-----")
    img_name=(imagestring + ".jpg")
    #cv2.imwrite(img_name,frame)/home/pi/tensorflow1/models/research/object_detection
    cv2.imwrite("/home/pi/Desktop/face_recognition/examples/High_tempdetected/"+img_name,frame)
else:
    espeak.synth("fever detected you cannot enter")
    imagestring=str(datetime.datetime.now().strftime("%I:%M:%S%p on %B %d,%Y"))

    text_file=open("nentry.txt","a+")
    n= text_file.write(name)
    n= text_file.write(temp)
    n= text_file.write(datetime.datetime.now().strftime("%I:%M:%S%p on %B %d,%Y"))
    n= text_file.write("-----")
    img_name=(imagestring + ".jpg")
    #cv2.imwrite(img_name,frame)
    cv2.imwrite("/home/pi/Desktop/face_recognition/examples/Normal_temp/"+img_name,frame)

else:
    print("old same")
```

Fig. 5: Code for saving the database

```
# Display the results
for (top, right, bottom, left), name in zip(face_locations, face_names):
    # Scale back up face locations since the frame we detected in was scaled to 1/4 size
    top *= 4
    right *= 4
    bottom *= 4
    left *= 4
    if bodytemp <= 37:
        cv2.rectangle(frame, (left, top), (right, bottom), (0, 128, 0), 2)
        cv2.rectangle(frame, (left, bottom - 35), (right, bottom), (0, 128, 0), cv2.FILLED)
        font = cv2.FONT_HERSHEY_DUPLEX
        cv2.putText(frame, name, (left + 6, bottom - 6), font, 1.0, (255, 255, 255), 1)
        cv2.putText(frame, temp, (left - 30, top - 26), font, 1.0, (0, 128, 0), 1)
    else:
        # Draw a box around the face
        cv2.rectangle(frame, (left, top), (right, bottom), (0, 0, 255), 2)
        cv2.rectangle(frame, (left, bottom - 35), (right, bottom), (0, 0, 255), cv2.FILLED)
        font = cv2.FONT_HERSHEY_DUPLEX
        cv2.putText(frame, name, (left + 6, bottom - 6), font, 1.0, (255, 255, 255), 1)
        cv2.putText(frame, temp, (left - 30, top - 26), font, 1.0, (0, 0, 255), 1)

# Display the resulting image
cv2.imshow('Video', frame)

# Mit 'q' on the keyboard to quit!
if cv2.waitKey(1) & 0xFF == ord('q'):
    break
```

Fig. 6: Code that shows the image along with the name and temperature

bian installed. It generally comes pre-installed with Python3 but if you do not find it, install the latest Python environment and IDE as well.

Open the Linux terminal and install the following libraries using the command given below (see Fig. 1):

```
sudo pip3 install face-recognition
sudo apt-get install espeak
sudo apt-get install espeak python-espeak
sudo pip3 install opencv2
```

Coding

After installing the libraries, open the Python IDE and start writing the code by importing the following libraries to it:

Espeak. To announce the temperature and name of the detected face.

Face_recognition. To recognise the face.

Date and time. To add the date and time of a person entering an area and later store the information in the database.

Opencv. To capture the video and process it frame-by-frame for facial recognition.

Next, set the image file name of each person we want to be recognised. Of course, we need to create

Connection of the MLX Sensor

Table 2

SDA	RPI SDA
SCL	RPI SCL
GND	RPI GND
VCC	RPI 3V



Fig. 7: Author's prototype